

Twitter/X Analytics Project

Engagement Analysis, Time-Based Patterns & Insights

Executive Summary

This project analyzes a Twitter/X analytics export of 1,166 valid tweets posted between June 1 and October 19, 2020 to evaluate reach, engagement behavior, top-performing content, and posting-time patterns. Python (Pandas) was used for data loading, quality checks, cleaning, feature engineering, and exploratory analysis. The central finding is that performance is highly skewed: a typical tweet earns ~299 impressions and 6 engagements, while a single tweet on July 10 generated ~17% of all impressions and ~59% of all engagements.

KPI Summary

KPI	Result
Total Tweets Analyzed	1,166 (after removing 15 invalid records)
Analysis Period	Jun 1, 2020 – Oct 19, 2020 (UTC)
Total Impressions	~911.7K
Total Engagements	~92.2K
Average / Median Impressions per Tweet	~781.9 / 298.5
Average / Median Engagements per Tweet	~79.1 / 6
Average / Median Engagement Rate	~3.64% / ~2.33%
Total Likes	7,883
Top Tweet (Jul 10, 2020)	156,685 impressions; 54,098 engagements

1. Project Objective

Transform raw Twitter/X performance data into actionable marketing insights on reach and impressions, engagement performance, content performance, and time-based performance patterns, to support future content planning.

2. Business Questions

Reach: how many impressions do tweets earn, and how are they distributed? Engagement: which interaction types drive engagement? Content: which tweets lead on impressions, engagements, and engagement rate? Timing: how do month, day of week, and hour of day affect performance? Relationships: how do impressions, engagements, and engagement rate move together?

3. Data Preparation and Methodology

Workflow: Raw Tweet Export (Excel) > Data Quality Checks > Data Cleaning > Feature Engineering > Exploratory Analysis > Business Insights. The raw file had 1,181 rows and 21 columns, with no duplicate rows or duplicate IDs. The time field was converted to a UTC datetime; 15 records could not be parsed and also carried inconsistent metrics (such as fractional impressions and engagement rates above 100%), so they were removed, leaving 1,166 records with no missing values. Seven columns (detail expands, permalink clicks, app opens, app installs, follows, email tweet, dial phone) were dropped from the core analysis, leaving 14 columns. Date, month, day of week, and hour features were derived, and the cleaned dataset was saved as a CSV.

4. Dataset Overview

Each record is one tweet with its timestamp, impressions, engagements, engagement rate, and six interaction measures: likes, retweets, replies, user profile clicks, URL clicks, and hashtag clicks. Media views and media engagements are retained in the cleaned file. A calculated-interactions field (sum of the six interaction measures) was created to compare against the reported engagements figure.

5. Reach and Visibility

Reach is concentrated in the low hundreds: 618 tweets (53%) earned 101–500 impressions, 146 earned 100 or fewer, and 210 earned 501–1K. Only 179 tweets reached 1K–5K, 10 reached 5K–10K, and just 3 exceeded 10K. The mean (~782) is well above the median (~299), confirming a long right tail driven by a few viral tweets.

6. Engagement Analysis

Engagement is similarly skewed. 542 tweets (46.5%) earned 0–5 engagements and 755 (64.8%) earned 10 or fewer, while only 9 tweets exceeded 500. Among the six measured interaction types, likes dominate.

Interaction	Total	Share of Interactions	Median per Tweet
Likes	7,883	51.9%	2
URL Clicks	2,829	18.6%	0
User Profile Clicks	2,568	16.9%	0
Replies	1,127	7.4%	0
Retweets	471	3.1%	0
Hashtag Clicks	303	2.0%	0

Reported engagements exceed the six-interaction sum for most tweets (median gap of 2), and the gap is largest on viral tweets (about 50.5K on the top tweet), indicating the engagements metric includes actions beyond those six types.

7. Top-Performing Tweets

The July 10, 2020 tweet is the clear outlier: 156,685 impressions, 54,098 engagements, and a 34.5% engagement rate. It alone accounts for ~17.2% of total impressions and ~58.6% of total engagements. Other leaders by impressions were posted on Sep 16 (18,501), Jun 26 (11,496), Jul 1 (9,820), and Aug 19 (9,743). The highest engagement rates came from different tweets: 39.0% (277 impressions, Jun 17) and 36.4% (1,271 impressions, Oct 6), showing that highly engaging content is not always the most widely seen.

8. Time-Based Performance

By month, July had the most tweets (312) and by far the most impressions (353.7K) and engagements (64.9K) because of the July 10 tweet; excluding it, July totals fall to ~197K impressions and ~10.8K engagements. Average engagement rate was highest in October (3.86%) and September (3.85%).

Month	Tweets	Impressions	Engagements	Avg Eng. Rate
June	223	132,703	5,876	3.46%
July	312	353,720	64,880	3.60%
August	218	149,361	7,079	3.47%
September	232	175,486	8,820	3.85%
October (to Oct 19)	181	100,428	5,589	3.86%

By day of week, Friday shows the highest average impressions (~1,638) and engagements (~364) per tweet, but this is driven by the same viral tweet; excluding it, Friday averages ~692 impressions and ~36 engagements, in line with other weekdays. Sunday has the highest average engagement rate (5.00%) but on only 80 tweets, and weekends together account for just 11% of posts.

Day	Tweets	Avg Impr./Tweet	Avg Eng./Tweet	Avg Eng. Rate
Monday	187	767.9	37.3	3.22%
Tuesday	239	555.1	27.1	3.44%
Wednesday	215	745.8	46.4	3.53%
Thursday	230	504.9	19.4	3.71%
Friday	165	1,637.9	363.8	3.90%
Saturday	50	733.8	30.4	3.24%
Sunday	80	650.2	34.8	5.00%

By hour (UTC), posting is concentrated between 12:00 and 19:59, which holds 741 tweets (63.6%), with the busiest hour at 17:00 (120 tweets). No tweets were posted between 05:00 and 07:59. The 13:00 hour leads on impressions (215K) and engagements (56K) only because of the viral tweet; excluding it, that hour totals ~58K impressions and ~2K engagements. Average engagement rate is highest at 10:00 (5.80%, 25 tweets) and around midnight to 01:00 (~4.7%), and lowest during the busiest afternoon hours (about 2.9%–3.0% at 12:00–15:00).

9. Correlation Analysis

Pearson correlation between impressions and engagements is very high (0.98), but this is inflated by extreme outliers; the rank-based Spearman correlation is a more moderate 0.72. Engagement rate has almost no rank relationship with impressions (Spearman 0.06) but a strong one with engagements (0.67). Grouping tweets into impression quartiles shows a non-linear pattern: the top quartile (above ~711 impressions) has the highest average engagement rate (5.21%; median 3.15%), the lowest-reach quartile is next (3.77%; median 2.70%), and the middle quartiles are lowest (2.65%–2.91%).

10. Key Business Insights

Performance is dominated by outliers: the mean is far above the median for impressions, engagements, and engagement rate. A single tweet drives most monthly and Friday results, so averages and totals should be read alongside medians. Likes account for over half of measured interactions, while retweets (3.1%) are the weakest amplification signal. URL and profile clicks together make up ~35% of interactions, showing meaningful traffic-driving behavior. High reach does not guarantee high engagement rate, and the best engagement-rate tweets had modest reach. Posting volume clusters in afternoon UTC hours, where engagement rates are lowest, while less crowded slots (10:00, late night, Sunday) show higher rates on smaller samples.

11. Business Recommendations

Study the content, format, and context of the breakout tweets (Jul 10, Sep 16, Jun 26, Jul 1) to identify repeatable elements. Report medians and outlier-adjusted figures alongside averages in performance reviews. Test posting at lower-volume slots such as 10:00 UTC, late evening/midnight, and Sundays, treating current results as hypotheses given small samples. Add calls to action that encourage retweets and sharing, and keep linking tweets to destinations, since URL and profile clicks are a strong secondary behavior. Monitor engagement rate and impressions together so that reach and resonance are both tracked.

12. Analysis Sections

1. Reach and Visibility 2. Engagement Analysis 3. Top-Performing Tweets 4. Time-Based Performance (Month, Day, Hour) 5. Correlation Analysis 6. Impression Quartile vs Engagement Rate

13. Conclusion

The project demonstrates an end-to-end analytics workflow from raw data validation and cleaning to feature engineering, distribution analysis, time-based analysis, and correlation analysis in Python. Major opportunities are replicating breakout content, improving amplification (retweets), optimizing posting times through structured testing, and reporting with outlier-aware metrics. Hashtag, mention, and media-type effectiveness, listed in the original objective, are natural next steps once tweet text and media type fields are available for analysis.

Data note: The tweet text in the source file is placeholder text, so content-level analysis (topics, hashtags, mentions) was not possible. All times are in UTC, October 2020 covers data only through Oct 19, and findings on timing are based on small samples for several hours and days.